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The power of digital nativeness: Exploring how millennials mitigate psychic distance in cross-border electronic commerce

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ABSTRACT

CBEC has experienced significant growth, establishing itself as an important component of international trade. Behind this expansion, digitally native millennial consumers have been one of the driving forces. However, existing literature lacks a comprehensive understanding of the ways millennials leverage their digital traits in CBEC. To address this gap, we explore how millennials enhance their purchase intentions in CBEC through the novel concept of digital nativeness. Our research model, supported by generation cohort theory and construal level theory, was tested on 298 South Korean samples using partial least squares structural equation modelling (PLS SEM). Our study identifies digital nativeness as a key driver of digital engagement, which reduces psychic distance and enhance exploratory purchasing in CBEC. We also found that social and transactional distance significantly impact exploratory purchasing, while spatial and temporal distance do not. Theoretically, this paper outlines information power concept that highlights millennials' influence on CBEC growth. It also tackles the psychic distance paradox by positioning CBEC platforms as strategic tools designed for millennials. Managerially, it emphasizes to target millennials with tailored strategies based on prioritised psychic distance dimensions. It also suggests a harmonized global-local approach to enhance millennials' engagement in CBEC.

1. Introduction

The emergence of Cross-Border Electronic Commerce (CBEC) – the practice of selling goods directly to foreign consumers through a digital intermediary – is regarded as one of the most notable shifts in the international business domain (Chen et al., 2023; Han & Kim, 2019; Yan et al., 2023). Large and well-known platforms, such as Amazon, Zalando and ASOS from the West are investing substantial resources to attract international customers. In the Asian region, Lazada, Shopee, Tmall Global, and Alibaba are acting as important intermediaries that link international merchants and consumers. Amazon leads the market with a total sales transaction value of US$360.4 billion in 2023, with Japan and the UK being its leading international markets. Shopee recorded a total sales transaction value of US$57.6 billion, with its primary international markets concentrated in the Asia Pacific region, including China, Vietnam, and Thailand (ECDB, 2024). From 2015 to 2019, CBEC maintained its growth rate from 7.4 % to 14.1 % (Yan et al., 2023). During the Covid-19 pandemic, CBECs volume surged by 25 % from 2020 to 2022. CBEC already accounts for 30 % of global e-commerce, with 60 % of these transactions being intercontinental (McKinsey, 2022). The global CBEC market is projected to grow at an impressive compound annual growth rate (CAGR) of over +21.6 % between 2024 and 2029 (Market Data Forecast, 2024). CBEC expected to reach 25 billion and 30 billion transactions by 2030 (McKinsey, 2022). This growth is solidifying its role as another pillar of international trade (Li & He, 2024).

The growth of CBEC has been made possible not only by digital platforms but also by the active role of consumers. Especially, millennial consumers (the generation born from 1980 through early 2000) (Moore, 2012; Purani et al., 2019) have been a driving force behind this growth (Forbes, 2021; Statista, 2024). Limited store access during the pandemic boosted millennials' digital purchasing (Giuffrida et al., 2021). The extensive variety of products on CBEC platforms has allowed millennials to leverage their unprecedented purchasing power (Li & He, 2024). Their digitally oriented consumption patterns have broadened the range of items in the CBEC market (Han & Kim, 2019). Millennials are at the forefront of ethical and environmentally friendly purchasing (Lavuri, 2022; Luger et al., 2022), even within CBEC transactions (Chatzopoulou & de Kiewiet, 2021; Cosgrave & O'Dwyer, 2020). Millennials are, thus worth to be regarded as the enabler of CBEC's recent dramatic growth.

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However, our understanding of millennials in CBEC growth remains incomplete, particularly empirically. While previous research has explored consumer purchase intentions, millennials as a distinct cohort have received limited attention. Specifically, most studies have failed to explicitly capture the inherent digital traits of millennials. Instead, scholars often described these traits using the term 'digital native' which refers to a digital generation (Purani et al., 2019) rather than specific traits. This highlights a research gap, namely the missing link between millennials' digital traits and purchase intention in CBEC.

To address this gap, we propose several novel approaches. First, we use the newly emerged concept of digital nativeness (Han, 2023; Hoffmann et al., 2014; Stokburger-Sauer & Plank, 2014, pp. 1–17) to capture millennials' inherent ease and positive attitude toward digital technology. This provides a clear and coherent basis to capture millennials' digital traits. Second, we argue that digital nativeness reduces psychic distance. While many risks in CBEC are seen as obstacles, existing research has focused on technological solutions to mitigate these risks from a seller's perspective. In contrast, we consolidate these risks into the concept of psychic distance, which millennials can reduce using their own digital traits. This approach shifts CBEC to a consumer-centric interface, where the role of digital traits can be tested effectively. Third, we argue that reduced psychic distance enhances millennials' exploratory buying behaviour, which in turn increase purchasing intention. Exploratory buying behaviour involves diverse and varied purchasing patterns (Waheed et al., 2018), making it suitable to examine whether millennials tend to engage in variety seeking behaviour, even in CBEC.

In summary, this study addresses the research question: How does digital nativeness impact the purchasing process of millennials in CBEC, particularly by reducing psychic distance in international markets? Given the importance of millennials in CBEC growth, this paper is one of the first to empirically examine how millennials develop their purchase intention from a consumer-centric perspective.

This study is expected to offer significant contributions. It may cultivate a clear and deeper understanding of how millennials' digital traits shape consumer engagement and drive the adoption of global e-commerce platforms. It also provides theoretical insights into consumer digital power in the evolving digital era. Additionally, it will offer practical guidelines for CBEC firms to design more effective strategies tailored to digitally native consumers.

This paper is structured as follows: Section 2 discusses related literature and identifies the research gap. Section 3 presents our research framework with key variables and outlines our hypotheses. Section 4 uncover the research method, followed by Section 5, which offers empirical test results. Section 6 provides discussions on the findings, implications, limitations, and future research directions.

2. Literature review

The recent advancements in CBEC research highlight its interdisciplinary evolution. Technological innovations focused on process improvement, such as blockchain for traceability (Liu & Li, 2020), artificial intelligence for price setting (Guo, 2022). From a supply chain perspective, studies highlight the importance of resource integration (Wang et al., 2020), risk management (Giuffrida et al., 2020) for CBEC firm performance. Regulatory developments like the EU's Digital Single Market and China's CBEC pilot zones prompted investigations the impact of EU Tax reforms on sellers (Fang & Ma, 2024) and CBEC firm' internationalization strategies (Cha et al., 2023).

Consumer behaviour is another critical dimension. Chen et al. (2023) proposed a framework that outlines political (e.g., policies), economic (e.g., economic distance), and sociocultural (e.g., social norms) stimuli that drives CBEC participation. Scholars highlight consumer benefits, such as access to diverse and unique foreign products (Huang & Chang, 2019; Makri et al., 2019; Mou et al., 2019). Such benefits have fostered research on motivations (Pop et al., 2023), loyalty (Li et al., 2024), trust (Huang & Chang, 2019). Social media transformed the way consumers enhance their value in different types in CBEC (Xu et al., 2023). Consumer value now includes socio-economic aspects such as purchasing from small foreign sellers (Wistedt, 2024), corporate social responsibility, and ethical consumption (Chatzopoulou & de Kiewiet, 2021; Cosgrave & O'Dwyer, 2020b). This growing literature highlights the evolving CBEC research that integrates technological, regulatory, and behavioural dimensions.

Behind the rapid growth of CBEC, millennials have been a key driving force (Forbes, 2021; Statista, 2024). Unlike digital immigrants who adopt technology later, millennials grew up immersed in digital technologies and naturally integrate digital tools into their lives (Bilgihan, 2016). This innate familiarity makes them both digitally fluent and innovative (Lissitsa & Kol, 2016; Purani et al., 2019). Their active information-seeking nature helps them navigate and leverage vast amounts of shopping information (Han & Kim, 2019). This promoted the use of social networks where they share shopping experiences (Kesharwani, 2020a; Xu et al., 2023). So, millennials are closely tied to social interactions, requiring diverse digital communication channels (Makri et al., 2019). Millennials differ from previous generations in three key ways: 1) their digital traits are inherent (nature) rather than acquired (nurture), leading to a familiarity with technology (Hoffmann et al., 2014; Roy et al., 2023); 2) they are active information seekers, rather than passive consumers (Han & Kim, 2019; Purani et al., 2019); and 3) they are goal-oriented in leveraging digital tools (Cosgrave & O'Dwyer, 2020b; Johnson et al., 2023). These digital traits align with the exploratory characteristics of CBEC (Mou et al., 2019), making it is reasonable to explore how millennials' digital traits influence their purchasing behaviour.

However, current literature lacks in-depth studies on the link between millennials' digital traits and purchase intention. This gap arises from reliance on conventional seller-oriented approaches that focus on external efforts to attract millennials. We define this as the dominant 'outside-in' approach.

The 'outside-in' approach is a major research stream that prioritizes extrinsic/environmental factors, focusing on how digital platforms, technologies, and marketing strategies influence consumer behaviour. The out-side in approach primarily emphasizes the valence framework (Yang et al., 2015). Negative valence in CBEC transactions arises from risks like extended delivery times, complex return, tracking processes and product uncertainty (Giuffrida et al., 2021; Mou, Cohen, et al., 2020a,b). High perceived risk reduces online shipping inclination and increase uncertainty (Ma et al., 2022), thus diminishing pre- and post-CBEC purchase intention (Mou, Cohen, et al., 2020a,b). Positive valence comes from perceived benefits such as a variety of foreign goods at reasonable prices, social value, and the adventure of exploration (Han & Kim, 2019). Consumers aim to maximize benefits while navigating these trade-offs (Mou, Cohen, et al., 2020a,b). Researchers have explored technical remedies to mitigate negative valence and enhance positive valence, focusing on how digital tools influence trust (Guo et al., 2018; Herrando et al., 2019). Key factors include platform quality (Cui et al., 2019), website design and policies (Huang & Chang, 2017, pp. 3981–3990). Perceived usefulness and ease of use also guide trust cultivation (Wistedt, 2024). Trust enhancement also encompass system-dependant factors like trust certification, encryption (Yang et al., 2015), third-party endorsements and reciprocal information exchange between buyers and sellers (Hoffmann et al., 2014).

It should be noted that the outside-in approach is now seen as a traditional method which may not valid to millennials who value proactive engagement and digital adaptability (Chen et al., 2023; Dash et al., 2021; Melović et al., 2021). In response, some studies proposed customer-based approaches. We define this as the 'inside-out' approach that focus on the intrinsic consumer attributes. This approach consider that consumption originates from the customer, and so focuses on the individual consumer's needs, preferences, and experiences, using insights from newly captured customer behaviours (e.g, Roy et al., 2023;

Wistedt, 2024). We interpret that inside-out approach examines the cognitive processes involved in online shopping, such as decision-making processes based on social identity (Shao et al., 2021). Moreover, it focuses on how consumers gather and evaluate information from various digital sources, such as information seeking (Mou, Cohen, et al., 2020a,b). Additionally, it looks at intrinsic motivations and preferences that influence consumer behaviour (Xu et al., 2023). This approach leads to some useful strategies. One proposed solution is to elevate the level of transaction and product information available to customers to offer abundant information as a source of shopping inspiration. This incorporates purchasing reviews (Valarezo et al., 2018), comments and style guides along with discussions in social media (Parker & Wang, 2016; Pop et al., 2023). Active product information disclosure also enhances cognitive involvement of the customers (Mou, Cohen, et al., 2020a,b; Zhu et al., 2019). Information signal including vendor reputation (Huang & Chang, 2017, pp. 3981–3990) boost purchase intentions of millennials. Personalised advertising to is critical to increase purchase intention (Smith, 2019). According to Makri et al. (2019) global and local consumer identity support purchasing intentions.

This inside-out approach seems to provide a suitable framework to include millennials' digital traits. However, in this stream, few studies focus on millennials as a target consumer group. Instead, previous research treats digital traits as a general consumer background in examining purchase intention (e.g., Li & He, 2024; Shao et al., 2021; Xu et al., 2023).

To sum up, CBEC research has made significantly progress. However, it still lacks a clear understanding of how to engage digital millennials more effectively. The prevalent "outside-in" approach is insufficient to digitally native millennials. Whie the "inside-out" approach, which emphasizes consumers' intrinsic behaviour appears more suitable, current CBEC research has yet to apply this perspective to millennials. In other words, there is limited investigation into the specific role of millennials in CBEC growth. In specific, insufficient focus on this cohort has left a missing link between their digital traits and purchase intention. This missing link is highlighted as a critical but a "discounted factor" highly worth investigating (Chen et al., 2023, p. 2626). This is particularly relevant in the continuously evolving digital era, where control increasingly shifts from the market to the consumer (Dash et al., 2021; Melović et al., 2021).

3. Theoretical framework and hypotheses development

3.1. Theoretical framework

Theories serve as the foundation to explain business phenomena, in turn the phenomena contributes to the evolution of theories (Kumar, 2017). However, in many cases, existing theories fail to address newly emerging topics (Kumar, 2017; Parvatiyar & Sheth, 2021). In such cases, the extension and synthesis of theories from other disciplines can offer advanced theoretical bases (Parvatiyar & Sheth, 2021). Theory advancement can be achieved by: 1) a cross-disciplinary examination on the object; 2) synthesising and extending existing theories for broader applicability; 3) an evidence-based evaluation through replicated reviews (Varadarajan, 2020). Such synthesis has been applied in CBEC consumer research, such as combining the technology acceptance model and commitment trust theory (Wistedt, 2024), signalling theory and attachment Theory (Huang & Chang, 2019), and commitment -trust theory and construal level theory (Cui et al., 2020). This paper faces a similar situation as it aims to capture newly emerged digital traits of millennials in CBEC which require theoretical reliability. In this vein, this paper uses Generational Cohort Theory to frame digital millennials within the CBEC context. It also employs Construal Level Theory to link the millennial consumers to psychic distance and exploratory buying behaviour (Fig. 1). Generation cohort theory argues that specific generation share life experience so develop homogeneous attitude to a specific subject such as economic opportunity, social values and shopping patterns (Bilgihan, 2016; Licsandru & Cui, 2019; Lissitsa & Kol,

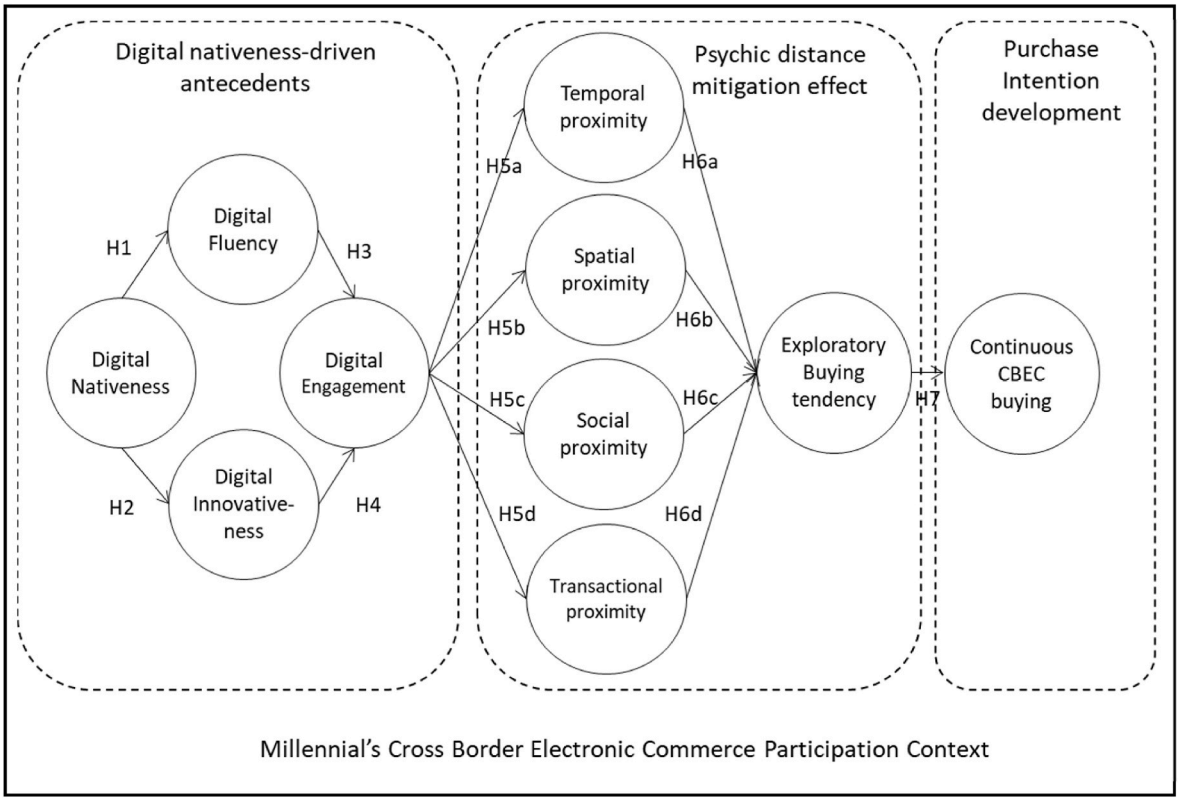


Fig. 1. Research model.

2016). Millennials, having been extensively exposed to digital media and devices from an early age, develop digital nativeness. This digital nativeness, in turn, enables them to activate other digital-focused attributes, namely digital fluency, digital innovativeness, and digital engagement. This approach, focusing on digital nativeness-driven antecedents, enabled us to examine how millennials manage international shopping in a more integrative manner that encompasses not only their ease with technology but also their ability to mitigate cognitive and physical barriers in CBEC. On the other hand, construal level theory justifies the existence of psychic distance in CBEC, thereby supporting the link between digital attributes and psychic distance. Construal theory asserts that individuals create mental representations of objects based on the psychic distance they perceive between themselves and those objects (Cui et al., 2020). In many cases, consumers perceive this distance when the object is not directly experienced and when there is also physical distance. This is consumer-oriented interpretation of 'transaction space' that has n-dimensional space defining international differences that must be accommodated in international transactions (Wood & Parr, 2006). Psychic distance encompasses four dimensions: temporal (ranging from earlier to later), spatial (ranging from closer to farther), social (ranging from familiar to unfamiliar), and hypothetical (ranging from certain to uncertain) (Cui et al., 2020). Following prior studies and considering CBEC process where the physical stores are not presented in the CBEC transactions, hypothetical distance is not examined (Cui et al., 2020; Darke et al., 2016; Han, 2022). Instead, we additionally investigate whether millennials reduce transactional distance. Transactional distance, in this context, represent the cognitive gap caused by varying technical and institutional environments in online interactions (Kostova et al., 2020; Lorentz et al., 2012). This distance is critical to understand how millennials interact with the sellers, as transactional risks disrupt the flow of information between sellers and customers (Ma et al., 2022).

We note that psychic distance highlights barriers that hinder consumer interactions with foreign markets. To focus on the positive role of digital nativeness, this paper shifts revert psychic distance to psychic proximity. This reframing illustrates how digital nativeness fosters digital interactions connectivity. The shift also provides an intuitive framework for understanding the relationships among key variables. This is particularly useful to capture the impact of digital engagement on distance mitigation and, in turn, the positive impact of distance mitigation on exploratory buying.

The hypothesis tests will first examine the impact of digital nativeness-driven attributes on psychic distance. Subsequently, they will explore the link between reduced psychic distance namely, psychic proximity and its effects on exploratory buying behaviour and continued CBEC buying intention. The latter two variables represent millennials' active and repeated involvement in CBEC. In the following sections, this paper delineates the definitions of the key variables and presents the arguments employed for the hypotheses within the research framework.

3.2. Hypothesis development

3.2.1. Digital nativeness and other digital traits

The concept of digital nativeness is relatively new. It stems from "digital native," a term widely used to specify the millennial generation in the digital age (Purani et al., 2019). Digital nativeness is defined as an individual's inherent ability to use digital technology with ease and efficiency, accompanied by a positive attitude toward digital technologies (Ebermann et al., 2016; Han, 2023; Stokburger-Sauer & Plank, 2014, pp. 1–17). This definition distinguishes digital nativeness from learned or acquired traits, such as technology adoption (Ariansyah et al., 2021) or digital trust building (Wistedt, 2024). It is regarded as a driver of digital conversion (Han, 2023), technology-task fit (Martins et al., 2023), and the social digital innovation adoption (Ebermann et al., 2016) so particularly useful to highlight millennials' inherent digital traits.

We propose that digital nativeness influences millennials' engagement with CBEC by driving two key traits: digital fluency and digital innovativeness. Digital fluency refers to the ability to use digital tools to find, create, and reformulate information in a digital environment (Wang et al., 2013). This concept goes beyond digital literacy for effective IT use as its focus is to leverage technology for different purposes especially in personal interactions and use experiences (Jarrahi & Eshraghi, 2019). Examples include the use of mobile technologies for multiple purposes (communication and socialising) in international shopping (Cui et al., 2020; Han & Kim, 2019) and social media use for both connection and learning (Jarrahi & Eshraghi, 2019). Digital innovativeness refers to the tendency to adopt new technologies, service, and experiences (Kim et al., 2010). Millennials are known for their innovative use of digital skills to discover new purchasing opportunities (Weber, 2017). Digital innovativeness is also regarded as the source of innovative product adoption (Zhang et al., 2024) and the utilisation of new digital service use, such as augmented reality (Wu et al., 2024). To represent engagement with CBEC, we incorporate digital engagement concept. Digital engagement refers to the level of involvement with digital interface (Dessart & Pitardi, 2019; Munaro et al., 2021). Digital engagement is different from digital experience, which is based on external stimuli (e.g., experiencing a brand) as the engagement requires voluntary behaviour (Dessart & Pitardi, 2019).

3.2.2. The impact of digital nativeness on digital fluency and digital innovativeness, which in turn both affect digital engagement

Regarding the impact on digital fluency, digital nativeness is positively associated with young consumer's fluent use of digital technologies (Han, 2023). Moreover, digitally native consumers are skilled in using CBEC technologies effectively (Han & Kim, 2019). In terms of the impact on digital innovativeness, digital nativeness is seen as an intrinsic ability that affects innovative IT use (Bunjak et al., 2024). Digital native consumers show a positive attitude toward innovative technology such as fast fashion mobile applications (Pop et al., 2023). The impact of digital fluency on digital engagement is well recognized. Consumers' confidence in using digital technology affect digital platform engagement (Roy et al., 2023). Young consumers' digital proficiency is crucial for fostering inclusive e-commerce engagement (Ariansyah et al., 2021). Millennials' ability to use platform functions is positively associated with both product and platform involvement (Xu et al., 2023). In terms of the impact of digital innovativeness on digital engagement, millennials' innovativeness is regarded as an antecedent to e-loyalty which is critical for consumer engagement (Purani et al., 2019). Digital nativeness, characterised by a positive attitude toward new technology, affects computer engagement, including control and compatibility (Kesharwani, 2020). Based on the previous arguments we set up hypothesis as below.

H1. Digital nativeness positively affects digital fluency

H2. Digital nativeness positively affects digital innovativeness

H3. Digital fluency positively affects digital engagement

H4. Digital innovativeness positively affects digital engagement

3.2.3. Psychic distance

Psychic distance refers to a perceptual remoteness between the home and foreign market, caused by different environmental factors such as language, business practice, and culture (Bandara et al., 2021). In applying this concept to the CBEC context, we take a more tailored approach to reflect the real situation. First, we include three commonly used subsets. Temporal distance refers to the time gap between buyer and seller in their interactions in a transaction process. Spatial distance denotes the physical separation between entities, ranging from proximity to remoteness. Social distance captures the level of unfamiliarity between individuals or groups (Cui et al., 2020; Hernández-Ortega, 2018). Given the complexity of CBEC, which introduces various

transaction risks (e.g., information sourcing, seller reputation, international logistics and payment), these three dimensions are not sufficient to cover CBEC barriers. To address this limitation, we propose a fourth dimension: transactional distance. Transactional distance is defined as the cognitive gap in communicating and conducting transactions in technical environments (Lorentz et al., 2012). While transaction risks are seen as critical barriers that require external aid (Ma et al., 2022; Mou, Cohen, et al., 2020a,b), how millennials overcome these risks has not been thoroughly examined. By introducing transactional distance, this paper explores the role of digital nativeness in CBEC transactions more comprehensively. In the context of online interaction, we define transactional distance as the cognitive gap caused by differing technical environments (Lorentz et al., 2012).

3.2.4. Impact of digital engagement on psychic distance

Previous literature hinted digital engagement reduces different types of psychic distances. Akram et al. (2022) indicates that a consumer's engagement in social networks effectively reduce the impact of spatial distance on network engagement. Uhm et al. (2022) showed that consumers with the digital interaction and realistic product information results in less social distance. Interactions to gain product information in international online purchasing is helpful in reducing social and transactional distance. Darke et al. (2016) revealed that spatial and social distance can be reduced when consumers are engaged with website tactics such as the owner information and image. Information acquired through electronic word of mouth including personal experiences, recommendations reduce the social psychological distance (Zhao et al., 2020). Platform interactions with customers in terms of the information quality decreases the social distance (brand familiarity and similarity) (Zhou & Jia, 2018). Temporal distance can be overridden with social network engagement (Hallam & Zanella, 2017). Cui et al. (2020) argued consumers engaged with mobile international purchasing also have reduced temporal distance. Based on this, this paper proposes hypotheses below.

H4a. Digital engagement of millennial consumers is positively associated to spatial proximity

H4b. Digital engagement of millennial consumers is positively associated to temporal proximity

H4c. Digital engagement of millennial consumers is positively associated to social proximity

H4d. Digital engagement of millennial consumers is positively associated to transactional proximity

3.2.5. Exploratory buying tendency

The exploratory purchasing behaviour refers to the consumer desire for exploration include risk taking in making purchase decisions (Waheed et al., 2018). This concept is not only about seeking different varieties of products but also information in terms of online purchasing so it has two different dimensions; information and product seeking (Waheed et al., 2018; Xu et al., 2023). In terms of e-commerce it is often translated into variety seeking behaviour (Li & Tuzhilin, 2023; Olsen et al., 2016).

3.2.6. Impact of psychic distance mitigation on exploratory buying tendency

Cui et al. (2020) identified that less temporal, geographical, social distance support the relationship commitment with active product offering information in CBEC so implied the increased level of exploratory buying behaviour. Cui et al. (2020) also viewed that reduced psychic distances actually enhance CBEC network connections. Close social psychological distance due to the interpersonal interactions in virtual interface affects the consumers to gain useful recommendations so decrease the source social distance (Yang, 2022). Based on this, this paper proposes hypotheses below.

H5. Spatial proximity of millennial consumers is positively associated to exploratory buying tendency

H6. Temporal proximity of millennial consumers is positively associated to exploratory buying tendency

H7. Social proximity of millennial consumers is positively associated to exploratory buying tendency

H8. Transactional proximity of millennial consumers is positively associated to exploratory buying tendency

3.2.7. Continuous CBEC intention

To highlight the impact of exploratory buying to consumer purchasing patterns, we chose purchase intention (Lude & Prügl, 2018). Purchase intention is a key outcome of consumer behaviour (Huang & Chang, 2019) and examined in the CBEC research(Chen & Yang, 2021; Shao et al., 2021). To encapsulate re-occurring purchasing patterns, we extend the concept to continuous purchase intention (Li et al., 2024; Mou, Cohen, et al., 2020a,b; Safeer et al., 2021). Continuous CBEC purchase intention concept covers not only an individual's willingness to transact online but also their product evaluation. Building on this, we define continuous CBEC purchase intention as customers' rational decisions based on acquired information which leads to repeated shopping patterns (Mou et al., 2019).

3.2.8. Impact of exploratory buying tendency on continuous CEBC intention

Consumers show higher purchase intention when more opportunities for information referral and time for navigation are provided (Xu et al., 2023). Information exploration, such as product description also enhances CBEC purchase intention (Mou, Cohen, et al., 2020a,b). Consumer informedness, acquired through details about price, process, and seller also enhance purchase intention in CBEC (Han & Kim, 2019). Based on this, we propose a hypothesis below.

H9. Exploratory buying is positively associated to continuous CBEC intention

4. Research method

4.1. Partial least squares structural equation modelling

This study uses PLS-SEM to test the model considering the objectives and characteristics of the research model. PLS-SEM is an effective technique for the early stages of theory-development and exploratory research that identifies relationships among variables (Hair, Hult, Ringle, Marko, 2013; Hair, Hult, Ringle, Sarstedt, 2017). This is because PLS-SEM aims to maximize the explained variance of the endogenous latent variables by estimating model relationships. In contrast, CB-SEM (covariance-based structural equation modelling) is recommended for theory testing and confirmative research, as it is a parameter-oriented method focusing on fit statistics to explain how well the data is explained in each research model (Dash & Paul, 2021; Hair et al., 2017, 2021; Rigdon, 2016; Sarstedt et al., 2016). The objective of this study encompasses both exploratory and theory-building approaches, PLS-SEM is more appropriate. In specific, this paper has an exploratory and predictive research context as the impact of digital nativeness and other digital-focused attributes on psychic distance are newly hypothesised. This study is also has theory-building approach as it newly theorizes the digital nativeness concept as a primary trait of millennial consumers in the specialized domain of CBEC. Following the recommendations of Hair et al. (2017), we use a two-stage PLS-SEM assessment method. The measurement models are first evaluated for each construct. Then, an assessment of the structural model with the hypotheses were conducted. SmartPLS 4.0 software was used for the two-stage analysis. As missing values were less than 5 % per indicator, they were addressed using mean value replacement. For the

bootstrapping procedure, 5000 subsamples from the original dataset were used to generate parameter coefficient estimates.

4.2. Measurement model operationalisation

The measurement indicators were adapted from prior relevant studies. Minor revisions were made to align them with CBEC research context. This approach is consistent with prior exploratory research (e.g., Wang et al., 2024; Wistedt, 2024; Xu et al., 2023). Given the exploratory nature of this study, some indicators were identified based on their alignment with definitional criteria. This approach is appropriate when widely accepted measurement scales are still under development, as is the case with recently emerging constructs such as millennials' digital traits (Kesharwani, 2020). For digital nativeness, we focused on its innate and inherent digital proficiency (DN1-4) based on the insights from Han (2023) and Teo (2013). Digital fluency was operationalized through efficiency and effectiveness in digital use skills (DF1−DF4), as ascertained by Jarrahi and Eshraghi (2019). Digital innovativeness indicators (DI1-4) were operationalized with its new and exploratory nature of digital use based on the work of (Kim et al., 2010). For digital engagement, the indicators (DE1-4) were drawn from Munaro et al. (2021) to reflect active participation in digital platforms. For proximity dimensions, four subsets of psychic proximity were measured. Temporal proximity (TP1-TP3) addressed the absence of time-related constraints in CBEC. Spatial proximity (SPP1-SPP3) focused on the irrelevance of physical distance in transactions. Social proximity (SOP1-SOP3) assessed trust and familiarity with international buyers and sellers. These three dimensions indicators followed Cui et al. (2020). Transactional proximity (TRP1-TRP3) evaluated cognitive ease in navigating technical and procedural aspects of CBEC, derived from Lorentz et al. (2012). For exploratory buying behaviour, we used indicators (EX1-EX4) that assess consumers' tendencies to explore different products and brand based on Waheed et al. (2018). Continuous CBEC purchase intention (CIB1-CIB4) measured repeated and ongoing engagement with CBEC platforms, adopted from Han and Kim (2019) (Mou, Cohen et al., 2020a). The scales were validated using the approach outlined by (Hair et al., 2017) Hair et al. (2017). This included assessing the measurement model for internal consistency reliability and validity, followed by evaluating composite reliability, factor loadings, average variance extracted, and discriminant validity. Appendix 1 contains the questionnaire used in this study.

4.3. Data collection and demographic profiles of respondents

In order to obtain reliable data for the study, we conducted a large-scale survey with the assistance of a leading research firm in South Korea. This firm manages a panel of 1.5 million participants, who are experienced consumers in various sectors such as e-commerce, consumer logistics, retail, and overseas purchasing. The large panel size is believed to include a substantial number of millennials. Moreover, we expect that many panel members would have experience with cross-border e-commerce (CBEC). We distributed an online questionnaire to randomly selected panel participants. To gather responses from millennial consumers with CBEC experience, we employed screening questions regarding the respondents' age and their experience with international purchasing from digital merchants. Furthermore, an additional formal check was conducted to verify their competency and adherence to the inclusion criteria. This check involved asking respondents about their familiarity and ability to navigate the CBEC process. Only respondents who fully answered these questions were included in the final data analysis. In summary, we implemented supplementary and structured methods to ensure that we collected data from millennial respondents who are knowledgeable about CBEC transactions. A total of 298 responses were collected. This approach is regarded as convenience sampling. This method is used in studies that require data with specific characteristics as is the case in this paper (i.e., millennials who have experienced CBEC transactions). While cconvenience sampling may limit the generalizability of findings, it is often the most practical and efficient method for accessing specific consumer groups when broader access is constrained. Prior CBEC studies (Chen & Yang, 2021; Han & Kim, 2019; Huang & Chang, 2019; Li et al., 2024) have demonstrated the effectiveness of this method in generating valuable insights. Table 1 summarizes the demographic profiles of the respondents.

In PLS SEM, determining the appropriate sample size is crucial for obtaining reliable results. We used the inverse square root method which uses the smallest significant path coefficient as proposed by Kock & Hadaya (2018). This method is recommended by Hair et al. (2021) with its strict approach, which contrasts with conventional calculations that tend to result in smaller sample size. For the calculation, the critical t-value corresponds to the desired significance level is divided by the smallest significant path coefficient in the model. The resulting value is then squared. In our case, at the significance level of 1 %, the critical t-value is known as 3.168. The smallest significant path coefficient in our test is 0.240 (H3 in Table 5). This gives us 3.168/0.240 = 13.2. The squared value of 13.2 is 174.24. This means that, for a 1 % significance level, at least 175 samples are required. Our sample size of 298 satisfies this requirement.

5. Empirical test and result

5.1. Common method bias test and non-response bias test

We first addressed the issue of common method bias, recognizing that the survey was completed by a single group of informants. To mitigate the potential for common method bias, we employed several procedural remedies (Kock et al., 2021), such as separating the dependent variables section in the questionnaire and using precise and clear wording. Additionally, we communicated assurances of confidentiality to the informants. To identify any presence of common method bias, we conducted Harman's single-factor analysis, which indicated that no single factor accounted for most of the variance. We also examined our variables for extremely high correlations (>0.90), which could suggest common method bias. As shown in Table 5, no such high correlations were found. To assess the possibility of nonresponse bias, we compared the responses from the first and last quartiles of respondents. Using the Mann–Whitney *U* test and the Wilcoxon matched-pairs signed-ranks test (Brace et al., 2012), we found no significant differences between these two groups ($p > 0.05$).

5.2. Measurement model analysis

We conducted four different types of validity tests for the measurement model analysis, as presented in Table 2. As suggested by Henseler et al. (2015), in PLS-SEM, composite reliability and Cronbach's alpha were used as thresholds for internal consistency reliability. Composite reliability and Cronbach's alpha values exceeding 0.60 were considered acceptable (Hair et al., 2017). In terms of composite reliability, the model's constructs scored between 0.807 and 0.899, and Cronbach's alpha scores ranged from 0.674 to 0.849. Therefore, these values satisfy the criteria. We assessed convergent validity using the average variance extracted (AVE), with a threshold of 0.50 or higher (Hair et al., 2017). Convergent validity refers to the extent to which a measure correlates with other measures of the same construct. The test results show that the AVE values met the required threshold.

Additionally, we examined indicator reliability, which looks at how much variation in an item is explained by the constructs. Outer loadings were used for this test, as they indicate an item's absolute contribution to its assigned variable. The required threshold for outer loadings is 0.708 or higher. Most of the indicators had outer loading values above 0.708, except for DF3 (0.647) and DL4 (0.696). In exploratory research, as is the case in this paper, a loading of 0.4 or higher can be considered

Table 1
Demographic profiles of the respondents (number of observations = 298).

Demographic Variables		Frequency	Ratio (%)	Demographic Variables	Frequency		Ratio (%)
Gender	Male	135	45	CBEC experiences (in years)	Less than 1 year	22	7
	Female	163	55		1–2 years	55	18
	Total	298	100		2–3 years	77	26
Age	22–24	18	6		3–4 years	53	18
	25–29	84	28		4–5 years	32	11
	30–34	112	38		More than 5 years	59	20
	35–39	62	21		Total	298	100
	40–41	22	7	CBEC experiences (in main items)	Baby suppliers	5	2
	Total	298	100		Fashion	108	36
Education	High school	47	16		Cosmetics	21	7
	University	236	79		Electronics	26	9
	Graduate school	15	5		Health supplements	81	27
	Total	298	100		Digital devices	38	13
Occupation	Office worker	169	57		Others	19	6
	Sales/Service	27	9		Total	298	100
	Technician	14	5	CBEC experiences (in expenses)	Under US $50	42	14
	Professional	25	8		US $51–$100	75	25
	Self-employed	12	4		US $101–$300	64	21
	Home wife	12	4		US $301–$500	63	21
	Others	39	13		US $501–$1000	21	7
	Total	298	100		US $1001–$2000	18	6
Household income per month	Less than US $1500	38	13		Over US $2000	15	5
	US $1500 - US $2300	99	33		Total	298	100
	US $2300 - US $3000	53	18				
	US $3000 - US $46000	31	10				
	Over US $46000	77	26				
	Total	298	100				

Source: Author

Table 2
Measurement model analysis.

Variables (abbreviations)	Number of Indicators	Internal Consistency Reliability		Convergent Validity	Indicator Reliability
		Composite Reliability	Cronbach's Alpha	AVE	Factor Loadings
Digital nativeness (DN)	4	0.889	0.833	0.667	0.741–0.847
Digital fluency (DF)	4	0.807	0.686	0.512	0.642–0.775
Digital innovativeness (DI)	4	0.894	0.843	0.679	0.795–0.875
Digital engagement (DE)	4	0.855	0.774	0.598	0.687–0.849
Temporal proximity (TP)	3	0.856	0.749	0.664	0.792–0.834
Spatial proximity (SPP)	3	0.888	0.811	0.727	0.795–0.895
Social proximity (SOP)	3	0.816	0.674	0.596	0.749–0.788
Transactional proximity (TRP)	3	0.877	0.792	0.704	0.781–0.869
Exploratory buying (EX)	4	0.859	0.781	0.604	0.752–0.795
Continuous CBEC use intention (CBI)	4	0.899	0.849	0.689	0.800–0.855

Source: Author.

satisfactory (Hulland, 1999). Appendix 2 presents the factor loadings for this analysis.

We assessed discriminant validity in two ways. First, the square root of the AVE for each construct should be higher than its highest correlation with any other construct. Table 3 shows that all values met this threshold according to the Fornell–Larcker criterion. Second, we

Table 3
Fornell–Larcker criterion analysis.

Latent Variables	CBI	DE	DF	DI	EX	SOP	SPP	TP	TRP	DN
CBI	0.830									
ED	0.617	0.773								
DF	0.541	0.503	0.715							
DI	0.615	0.663	0.480	0.824						
EX	0.560	0.536	0.482	0.545	0.777					
SOP	0.559	0.634	0.445	0.560	0.476	0.772				
SPP	0.628	0.384	0.441	0.462	0.468	0.376	0.853			
TP	0.611	0.391	0.490	0.433	0.484	0.404	0.768	0.815		
TRP	0.532	0.338	0.531	0.402	0.526	0.418	0.571	0.589	0.839	
DN	0.503	0.520	0.666	0.590	0.451	0.452	0.411	0.427	0.446	0.817

Note: Refer to Table 2 for abbreviations.
Source: Author.

examined cross-loadings. In this analysis, an indicator's outer loading should be higher than all of its cross-loadings with other constructs. Appendix 2 shows that our model satisfied the recommended cross-loading levels.

We also evaluated discriminant validity using the Heterotrait-Monotrait Ratio (HTMT) (Hair et al., 2017). Discriminant validity is confirmed when the HTMT value is below 0.90, indicating that the two reflective constructs are sufficiently distinct. The HTMT results in Table 4 demonstrate that most of the values fall within the acceptable range.

5.3. Structural model analysis

After confirming the validity of the measurement variables, we conducted the structural model analysis (Hair et al., 2013; 2017). An overview of the models analysed using PLS-SEM is presented in Table 5. The table includes the explained variance (R^2), path coefficients, and t-values. They are determined by bootstrapping to assess significance. The test result also highlights how digital nativeness, fluency, and engagement influence different types of psychic proximity and exploratory purchasing. Hypotheses H1a ($\beta = 0.666$, $p < 0.01$) and H1b ($\beta = 0.590$, $p < 0.01$) are supported, indicating that digital nativeness enhances these two factors. The results show that digital nativeness positively affects both digital fluency and digital innovativeness. For digital fluency and digital innovativeness on digital engagement, the findings support H3 ($\beta = 0.240$, $p < 0.01$) and H4 ($\beta = 0.547$, $p < 0.01$). This confirms that both factors significantly contribute to higher digital engagement. Digital engagement also plays a key role in improving various types of proximity. The analysis supports H5a ($\beta = 0.391$, $p < 0.01$), H5b ($\beta = 0.384$, $p < 0.01$), H5c ($\beta = 0.634$, $p < 0.01$), and H5d ($\beta = 0.338$, $p < 0.01$), indicating a positive impact on temporal, spatial, social, and transactional proximity.

However, the effects of psychic proximity on exploratory purchasing vary. Social proximity (H6c: $\beta = 0.267$, $p < 0.01$) and transactional proximity (H6d: $\beta = 0.275$, $p < 0.01$) positively influence exploratory purchasing. On the other hand, temporal (H6a: $\beta = 0.126$, nonsignificant) and spatial proximity (H6b: $\beta = 0.114$, nonsignificant) show no significant effects. Lastly, exploratory purchasing positively associated to the intention to continue using CBEC. Hypothesis H7 ($\beta = 0.560$, $p < 0.01$) is supported, indicating a clear link between exploratory purchasing and continuous CBEC use intention.

Overall, digital nativeness enhance digital fluency and digital innovativeness, which in turn increase digital engagement. Digital engagement enhances psychic proximity which represents reduced psychic distance. In particular, social and transactional proximity is identified as drivers of exploratory purchasing. This purchasing behaviour further strengthens the intention to continue intention to use CBEC (see Fig. 2).

Table 5
Effects and variance explained for endogenous variables.

Effects on Endogenous Variables with Hypotheses	Path Coefficient β (*t*-value)	Variance Explained (R^2)
		Supported or Not supported
Effects on Digital fluency	0.666	0.443
H1a: Digital nativeness → Digital fluency	(19.845***)	Supported
Effects on Digital innovativeness	0.590	0.348
H1b: Digital nativeness → Digital innovativeness	(14.283***)	Supported
Effects on Digital engagement		0.484
H3: Digital fluency → Digital engagement	0.240 (4.498***)	Supported
H4: Digital innovativeness → Digital engagement	0.547 (10.166***)	Supported
Effects on Temporal proximity	0.391	0.153
H5a: Digital engagement → Temporal proximity	(6.462***)	Supported
Effects on Spatial proximity	0.384	0.186
H5b: Digital engagement → Spatial proximity	(6.376***)	Supported
Effects on Social proximity	0.634	0.402
H5c: Digital engagement → Social proximity	(18.807***)	Supported
Effects on Transactional proximity	0.338 (6.143***)	0.114
H5d: Digital engagement → Transactional proximity		Supported
Effects on Exploratory purchasing		0.386
H6a: Temporal proximity → Exploratory purchasing	0.126 (1.406 *NS*)	*Not supported*
H6b: Spatial proximity → Exploratory purchasing	0.114 (1.435 *NS*)	*Not supported*
H6c: Social proximity → Exploratory purchasing	0.267 (4.624***)	Supported
H6d: Transactional proximity → Exploratory purchasing	0.275 (3.685***)	Supported
Effects on Continuous CBEC use intention	0.560 (12.240***)	0.313 Supported
H7: Exploratory purchasing → Continuous CBEC use intention		

Note: ***p < 0.01; **p < 0.05; *p < 0.1 (all two-tailed), NS: nonsignificant.
Source: Author.

6. Conclusion and discussions

6.1. Findings

Although millennial consumers have a driving force of CBEC growth, their roles have not been specifically examined within the context of CBEC. Naturally, there is a missing link between millennials' digital traits and their purchasing intentions. This paper addresses this gap using the novel concept of digital nativeness and identifies several findings.

First, this paper responds to Chen et al. (2023)'s call to identify the

Table 4
Heterotrait–Monotrait Ratio (HTMT).

Latent Variables	DE	DF	DI	EX	SOP	SPP	TP	TRP
DE	0.752							
DF	0.707	0.685						
DI	0.731	0.824	0.625					
EX	0.684	0.683	0.655	0.671				
SOP	0.713	0.818	0.647	0.723	0.644			
SPP	0.755	0.479	0.598	0.560	0.585	0.510		
TP	0.760	0.505	0.685	0.542	0.621	0.565	0.977	
TRP	0.644	0.412	0.733	0.482	0.650	0.552	0.713	0.757
DN	0.597	0.652	0.884	0.702	0.555	0.602	0.501	0.535

Note: Refer to Table 2 for abbreviations.
Source: Author.

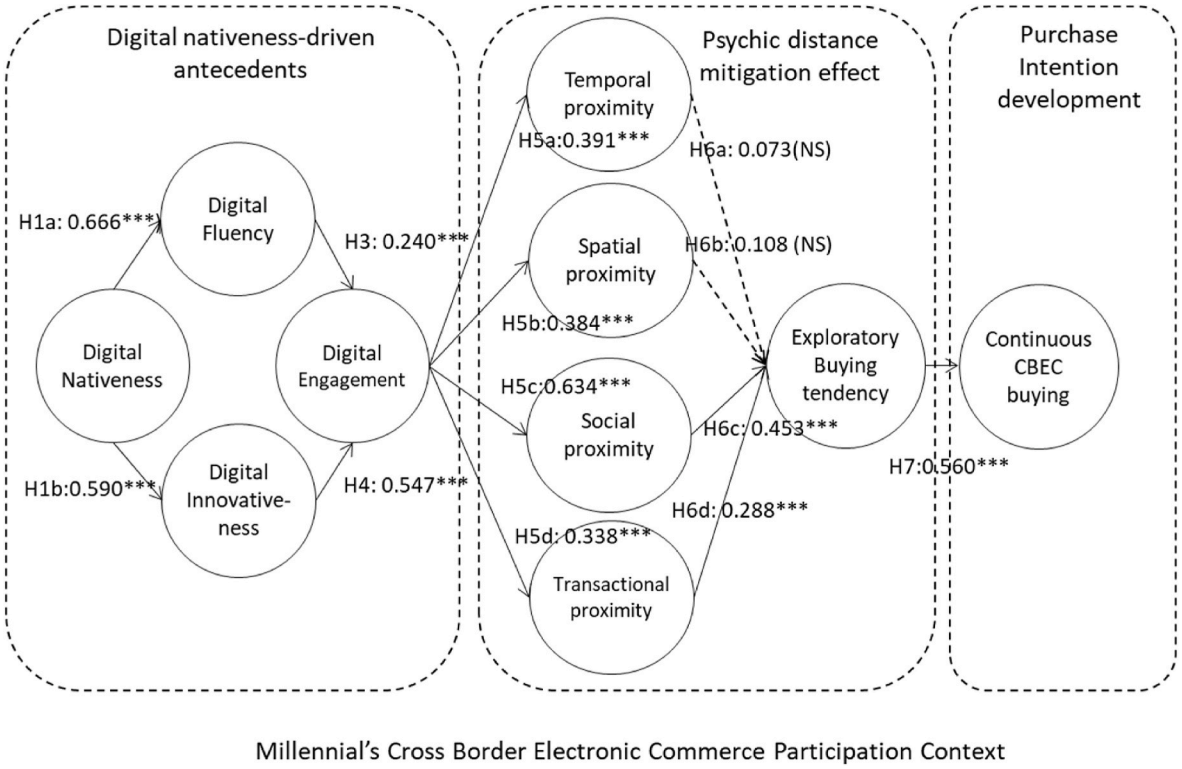


Fig. 2. Results of path analysis.
Note: ***p < 0.01; **p < 0.05; *p < 0.1 (all two-tailed), NS: nonsignificant.
Source: Author.

'unique characteristics' that drive consumers to engage in CBEC. From earlier research on millennials in online commerce (e.g., Smith, 2011) to recent studies in CBEC (Li & He, 2024; Mou, Cohen, et al., 2020a,b; Wistedt, 2024), millennials' digital traits remain unexplored. By framing digital platforms as the key mechanism for overcoming geographical barriers, this study reveals that digital nativeness is aligned with the evolution of digital commerce networks. Through this novel perspective, we define digital nativeness as not only a distinctive trait of millennials but also a driving factor behind their participation in international transactions.

Second, we establish that millennials' digital nativeness diminishes psychic distance. Due to the lack of clear connection between millennials and CBEC engagement, the way millennials promote their purchase intention has not been fully explained. By defining psychic distance as international barriers, this paper finds that millennials' digital traits enable them to reduce these barriers autonomously. This shifts perspectives in digital consumer research from an external aid-driven strategy to one focused on intrinsic consumer characteristics. This finding lays the groundwork for adapting the inside-out approach to better align with technology-use theoretical models, particularly in CBEC. This also offers guidance on future consumer behaviour research in the digital era (Malter et al., 2020) to consumer digital traits.

Third, this paper reveals that the mitigation of social and transactional distance positively affects millennials' exploratory purchasing behaviours. Therefore, our proposed theory, which links reduced psychic distance and variety-seeking, receives empirical support as previous studies (Cui et al., 2020). However, the reduction of temporal and spatial distance shows no significant impact on exploratory purchasing. We interpret that, this is because, digital native millennials view these constraints as baseline conditions rather than barriers. So, with established digital infrastructures and normalized expectations of delivery times and logistics, temporal and spatial factors are less salient in driving exploratory behaviour. Instead, millennials focus on social and transactional proximities, such as trust in foreign sellers and seamless payment processes, which more directly enhance their shopping experience. In other words, exploratory purchasing is influenced more by experiential factors than practical constraints like time or space.

6.2. *Theoretical and managerial implications*

This paper offers several theoretical implications. First, it enriches the concept of consumer power in digital era (Hsieh et al., 2022). Based on our empirical result, we offer another type of power, namely information power rooted in digital nativeness. Millennials increasingly assert their influence in the global market, not only through their purchasing power but also through their digital capabilities. Consumer power has evolved significantly in the digital age, as digital consumers with digital nativeness are more actively engaged in cross-border shopping. This research highlights how digital nativeness has become a pivotal factor in shaping consumer power in the digital era.

Second, this study provides insights into e-commerce globalization from the customer's perspective. Globalization is often seen as a "state of mind" which reflects the various stances consumers adopt toward global engagement (Salnikova & Grunert, 2020; Vătămănescu et al., 2020). We propose that globalization, viewed through the lens of digital nativeness, can be described a "state of digital skills" that enables consumers to adopt technological approaches to international purchasing. This finding also offers valuable insights into the recent millennials' value seeking behaviour. Digital nativeness enhances millennials' digital skills to seek their social value they wish to pursue in the global digital marketplace (Cosgrave & O'Dwyer, 2020; Salnikova & Grunert, 2020).

Third, this paper offers a counter-perspective on the "psychic distance paradox" in online internationalization (Magnusson et al., 2014). Like international consumers, firm managers face psychic distance during online internationalization. The "psychic distance paradox" suggests that firms often perform better in distant markets than in

proximate ones due to an overly-reliance on virtual interactions with customers in limited physical presence. While the physical seller images remain important in international retailing (Darke et al., 2016; Wistedt, 2024), we can consider the role of digital-native millennials, who autonomously reduce psychic distance through their digital traits. This suggest that virtual interactions can be a strategic touchpoint for engaging customers, not a trap. This reframing shift focus from managerial adaptations to the active role of consumers in bridging market gaps.

We also offer several managerial implications. First, CBEC firms should segment millennials as a key consumer cohort. Moreover, they need to evaluate and leverage their digital nativeness to enhance exploratory purchasing. For instance, firms can employ behavioural analytics (e.g., data analytics or machine learning) to categorize consumers by their exploratory buying behaviour. For highly exploratory consumers, firms can tailor product offerings to include diverse, novel, and culturally rich products, aligning with their preferences for variety and unique experiences. This approach ensures a more targeted strategy that resonates the millennials' exploratory purchasing tendencies.

Second, firms should adopt an "inside-out" approach that aligns with millennials' digital traits. In doing so, our findings on psychic distance can guide CBEC platform design. Platforms should prioritize the four dimensions of psychic distance (i.e., temporal, spatial, social, and transactional) when designing their interface. Given the importance of social and transactional dimensions, platforms should enhance trust-building mechanisms (e.g., seller credentials) and facilitate seamless transactions (e.g., automated payment systems) to simulate social proximity. While spatial and temporal distances are less impactful for millennials, they remain fundamental. Therefore, platforms should focus on ensuring reliable logistics. By adopting this distance-based framework, CBEC platforms can effectively engage millennials and optimize their shopping experience.

Third, given the role of exploratory buying behaviour in increasing purchase intention, we observe that products, competitive prices, and unique items once considered exclusive are now more accessible through CBEC, fuelled by millennials' exploratory buying behaviour. This has blurred the distinction between domestic and international purchasing. Given this transition, CBEC business need to develop strategies that integrate global and local market approaches, allowing seamless transitions between domestic and international operations. This might include harmonizing product offerings and supply chains, pricing strategies, and promotional activities across borders.

6.3. Limitations and future research directions

We used data from a large panel pool in South Korea considering South Korea has a high CBEC penetration rate, and its consumers have extensive experience with CBEC transactions. This focus was due to the constraints of the survey conditions to collect data from millennials who experience CBEC. However, relying only on South Korean data may limit the generalizability of the findings. A large-scale survey across multiple countries could offer broader and more insightful results. Moreover, this paper focused on millennials using large and general CBEC platforms. There are platforms dedicated to specific area such as fashion or luxury goods. Furthe research can aim different types of platforms linked to digital nativeness and other millennials traits. Additionally, the role of millennials in newly emerging markets, such as those in developing countries or under-developed countries, should be recognized as a new opportunity for further CBEC expansion. In this vein, future research could focus on millennials in these developing regions to uncover unique dimensions and opportunities.

CRediT authorship contribution statement

Jeong Hugh HAN: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Po-Lin Lai:** Writing – review & editing, Visualization, Supervision, Conceptualization.

Appendix 1. Measurement indicators in questions

Indicators	Questions
DN1	I feel comfortable and natural using digital devices.
DN2	I am naturally well aware of the latest trends in digital devices.
DN3	I naturally excel in in using digital devices.
DN4	It is very easy for me to communicate through digital devices.
DF1	I can communicate effectively and meaningfully through digital devices.
DF2	I efficiently obtain lifestyle and purchasing information through digital tools.
DF3	I efficiently manage emails and messages through digital devices
DF4	I can use digital devices for multiple purposes (e.g., maps, navigation).
DI1	I always have great interest in new digital devices and technologies.
DI2	I actively seek and adopt new digital products, brands, and services.
DI3	I prefer various online platforms that provide the latest information.
DI4	I feel satisfaction in being an early adopter of unique and innovative digital products.
DE1	I actively share product information and shopping experiences.
DE2	I exchange information and share interests with people who have similar tastes, maintaining good relationships.
DE3	I gather necessary information from various communities via digital channels.
DE4	I directly participate in communication activities through various digital channels.
TP1	I can make international purchases at any time, without being limited by time.
TP2	I can easily check the order status in real-time when making international purchases via the internet.
TP3	The time zone difference between the seller and me does not significantly affect the purchasing process.
SPP1	I can make international purchases regardless of location.
SPP2	I can make international purchases while on the go.
SPP3	The physical distance between the seller and me is not important when making international purchases.
SOP1	I enjoy learning about foreign cultures and people.
SOP2	I believe that foreigners and international sellers can be trusted trading partners.
SOP3	I feel a sense of familiarity with other international shoppers like me.
TRP1	Having multiple payment options makes the international purchase process more convenient.
TRP2	Communication with sellers via chat, messages, cyber customer service, and FAQs helps with the purchasing process.
TRP3	Shipping prediction, customs calculation, and customs information are useful for making purchasing decisions.
EX1	I consider various international brands when making international purchases.
EX2	If the price and performance are similar, I look for alternative brands to the one I want.

(*continued on next page*)

(*continued*)

Indicators	Questions
EX3	I am willing to purchase alternative products if the price and performance are similar.
EX4	I gather and compare various product information when making international purchases.
CIB1	I will recommend international purchases to those around me.
CIB2	I will make repeat purchases through international buying.
CIB3	Continuous international buying is not difficult.
CIB4	I will continue to enjoy international buying.

Appendix 2. Cross loading analysis result

	DN	DF	DI	DE	TP	SPP	SOP	TRP	EX	CIB
DN1	**0.830**	0.482	0.436	0.344	0.360	0.366	0.338	0.448	0.339	0.362
DN2	**0.847**	0.482	0.563	0.459	0.332	0.322	0.396	0.321	0.409	0.413
DN3	**0.844**	0.491	0.561	0.496	0.317	0.328	0.460	0.300	0.347	0.424
DN4	**0.741**	0.701	0.361	0.388	0.381	0.325	0.279	0.394	0.373	0.435
DF1	0.468	**0.735**	0.365	0.447	0.335	0.263	0.327	0.300	0.343	0.421
DF2	0.502	**0.775**	0.422	0.352	0.468	0.411	0.396	0.526	0.426	0.494
DF3	0.423	**0.702**	0.365	0.448	0.248	0.251	0.347	0.292	0.317	0.317
DF4	0.524	**0.642**	0.202	0.161	0.353	0.345	0.185	0.411	0.288	0.303
DI1	0.527	0.348	**0.800**	0.507	0.343	0.414	0.381	0.349	0.459	0.475
DI2	0.514	0.431	**0.875**	0.563	0.312	0.324	0.473	0.290	0.479	0.464
DI3	0.518	0.485	**0.824**	0.519	0.446	0.469	0.501	0.470	0.497	0.580
DI4	0.380	0.314	**0.795**	0.598	0.326	0.316	0.491	0.213	0.356	0.511
DE1	0.419	0.432	0.538	**0.849**	0.362	0.356	0.567	0.325	0.493	0.555
DE2	0.389	0.363	0.540	**0.841**	0.328	0.309	0.618	0.296	0.460	0.562
DE3	0.428	0.396	0.399	**0.687**	0.239	0.279	0.364	0.240	0.322	0.366
DE4	0.390	0.377	0.573	**0.702**	0.266	0.233	0.368	0.166	0.357	0.390
TP1	0.396	0.440	0.392	0.339	**0.834**	0.710	0.322	0.565	0.453	0.552
TP2	0.328	0.391	0.339	0.289	**0.819**	0.568	0.304	0.454	0.345	0.463
TP3	0.312	0.361	0.321	0.323	**0.792**	0.584	0.362	0.408	0.372	0.471
SPP1	0.356	0.387	0.387	0.350	0.701	**0.865**	0.325	0.498	0.403	0.575
SPP2	0.360	0.394	0.414	0.362	0.646	**0.895**	0.324	0.483	0.387	0.545
SPP3	0.334	0.345	0.381	0.264	0.616	**0.795**	0.313	0.479	0.409	0.483
SOP1	0.370	0.365	0.408	0.406	0.325	0.296	**0.779**	0.298	0.371	0.398
SOP2	0.343	0.330	0.365	0.362	0.304	0.299	**0.749**	0.297	0.319	0.337
SOP3	0.341	0.340	0.496	0.637	0.310	0.283	**0.788**	0.360	0.399	0.519
TRP1	0.358	0.447	0.262	0.195	0.470	0.456	0.290	**0.781**	0.349	0.386
TRP2	0.398	0.458	0.384	0.316	0.516	0.488	0.381	**0.864**	0.478	0.488
TRP3	0.368	0.439	0.348	0.317	0.498	0.494	0.368	**0.869**	0.477	0.456
EX1	0.352	0.328	0.371	0.427	0.346	0.337	0.333	0.396	**0.752**	0.405
EX2	0.358	0.363	0.504	0.429	0.354	0.369	0.396	0.366	**0.778**	0.448
EX3	0.306	0.367	0.441	0.460	0.343	0.312	0.410	0.314	**0.795**	0.429
EX4	0.382	0.431	0.380	0.358	0.448	0.428	0.343	0.540	**0.783**	0.454
CIB1	0.464	0.447	0.536	0.584	0.449	0.459	0.510	0.444	0.425	**0.800**
CIB2	0.437	0.460	0.491	0.488	0.487	0.523	0.453	0.446	0.493	**0.855**
CIB3	0.409	0.461	0.472	0.440	0.615	0.605	0.437	0.504	0.467	**0.809**
CIB4	0.365	0.429	0.549	0.545	0.477	0.495	0.461	0.376	0.470	**0.855**

Data availability

The data that has been used is confidential.

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